

PCCST501 – COMPUTER NETWORKS

Assignment 3 | Module 3 – Data-Link Layer, Wireless LANs & Mobile IP

LAN, Ethernet & Wireless Network Investigation

Course Outcome: CO3 **Due Date:** 17 AUGUST 2026

Analyze how data-link protocols provide node-to-node delivery, framing, addressing and access to shared media, with emphasis on Ethernet and wireless networking.

Instructions

1. Answer all questions. 2. Use diagrams wherever appropriate. 3. Cite any external material used. 4. Submit the PDF report through the prescribed GitHub repository and Google Form. 5. Screenshots must be readable and relevant.

Assignment Questions

1. Explain the responsibilities of the data-link layer and distinguish logical link control from media access control.
2. Explain Ethernet frame format and identify the purpose of destination MAC, source MAC, type/length and FCS fields.
3. Use Wireshark to capture Ethernet traffic and identify source and destination MAC addresses from at least five frames.
4. Explain the operation of a switch using MAC address learning. Demonstrate how a switch builds its forwarding table.
5. Compare hub, switch and bridge in terms of forwarding, collision domains and practical use.
6. Explain CSMA/CD and discuss why its importance is reduced in modern switched full-duplex Ethernet.
7. Explain CSMA/CA in IEEE 802.11 and compare its objective with CSMA/CD.
8. Design a small wireless LAN for a college laboratory. Specify access point placement, SSID, channel considerations, addressing and basic security requirements.
9. Explain Mobile IP and the need for mobility support when a device moves between networks.
10. Perform a packet-level comparison of a wired Ethernet frame and a wireless LAN frame using suitable captures or documented examples.

Expected Deliverables

- PDF report
- Ethernet frame diagram
- Wireshark screenshots
- Wireless LAN design
- Comparison table
- README.md

Suggested Evaluation

Criteria	Marks
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Correctness & Technical Content	4
Documentation & Diagrams	2
GitHub Repository & Organization	2
Presentation / Viva	2
Total	10

Submission Note

Create or use the assignment repository provided by the course, include a clear README.md, organize evidence into folders where appropriate, and submit the repository link through the course submission form.